

09/04/20

S T Q Q S S D

$$\textcircled{C} \int_0^1 \frac{dx}{\sqrt{3x+1}} = \int_0^1 \frac{1}{\sqrt{3x+1}} dx = \int_0^1 \frac{1}{\sqrt{u}} \cdot \frac{1}{3} = \frac{1}{3} \int_0^1 \frac{1}{\sqrt{u}} = \frac{1}{3} \int_0^1 u^{-\frac{1}{2}} =$$

$$u = 3x+1 \rightarrow \frac{2}{3} u^{\frac{1}{2}} \bigg|_0^1$$

$$\frac{du}{dx} = 3 \rightarrow dx = \frac{1}{3} \text{ ou } \frac{2\sqrt{u}}{3}$$

$$\frac{2}{3} \sqrt{3x+1} \bigg|_0^1 = \frac{2}{3} \sqrt{4} - \frac{2}{3} \sqrt{1} = \frac{4}{3} - \frac{2}{3} = \frac{2}{3} \text{ u.a.}$$

③ Calcular a área definida pelas funções $f(x) = x^2 - 4$ e $g(x) = -x^2 + 2x$, nos intervalos definidos pela região sombreada da figura abaixo:

$$A = \left[\begin{array}{c} -2 \textcircled{A} \\ \int f(x) - \int g(x) \\ -3 \end{array} \right] + \left[\begin{array}{c} 1 \textcircled{C} \\ \int g(x) - \int f(x) \\ -2 \end{array} \right] \rightarrow$$

$$A = 7/3$$

$$B = 4/3$$

$$C = 0$$

$$D = -3$$

Fazendo integrais:

$$\int f(x) = \frac{x^3}{3} - 4x, \quad \int g(x) = -\frac{x^3}{3} + x^2$$

Calculando A, B, C, D.

$$A = [7/3 + 4/3] + [0 + 9]$$

$$A = 11/3 + 9 = 11/3 + 27/3$$

$$A = \frac{38}{3} \text{ u.a.}$$

$$\textcircled{A} = \frac{(-2)^3}{3} - 4 \cdot (-2) = -\frac{8}{3} + 8 = -\frac{8}{3} + \frac{24}{3} = \frac{16}{3}$$

$$\frac{16}{3} - 3 = \frac{16}{3} - \frac{9}{3} = \frac{7}{3}$$

$$\frac{(-3)^3}{3} - 4 \cdot (-3) = -\frac{27}{3} + 12 = -9 + 12 = 3$$

$$\textcircled{B} = -\frac{(-2)^3}{3} - (-2)^2 = \frac{8}{3} - 4 = \frac{8}{3} - \frac{12}{3} = -\frac{4}{3}$$

$$-\frac{(-3)^3}{3} - (-3)^2 = \frac{27}{3} - 9 = 9 - 9 = 0$$

$$\textcircled{C} = -\frac{1}{3} - 1 = -\frac{1}{3} - \frac{3}{3} = -\frac{4}{3}$$

$$\frac{8}{3} - 4 = \frac{8}{3} - \frac{12}{3} = -\frac{4}{3}$$

$$\textcircled{D} = \frac{1}{3} - 4 = \frac{1}{3} - \frac{12}{3} = -\frac{11}{3}$$

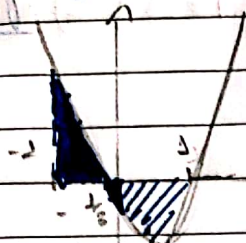
$$-\frac{8}{3} + 8 = -\frac{8}{3} + \frac{24}{3} = \frac{16}{3}$$

$$6 - 9 = -3$$

spiral

4) Achar a área da região limitada pela parábola $y = 3x^2 - 2x - 1$, pelo eixo "x" entre os pontos -1 e 0. Faça o gráfico.

$A_T = \int_{-1}^{-1/3} f(x) dx + \int_{-1/3}^0 f(x) dx$
 $\int f(x) = \int 3x^2 - 2x - 1 = x^3 - x^2 - x$



Calculando A:

$$\left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) = -\frac{1}{27} - \frac{1}{9} + \frac{1}{3} = -\frac{1}{27} - \frac{3}{27} + \frac{9}{27} = \frac{5}{27}$$

$$\left(-1\right)^3 - \left(-1\right)^2 - \left(-1\right) = -1 - 1 + 1 = -1 \quad \frac{5}{27} + 1 = \frac{5}{27} + \frac{27}{27} = \frac{32}{27}$$

Calculando B:

$$0 - \frac{0 - 5}{27} = \frac{5}{27} \quad A = \frac{32}{27} - \frac{5}{27} = \frac{27}{27} = 1 \text{ u.a.}$$

5) Calcule os integrais por parte: $\int \frac{\sin(5x)}{5} dx$

a) $\int x \cos(5x) dx = \frac{x \sin(5x)}{5} - \int \frac{\sin(5x)}{5} dx \rightarrow$

$$\frac{x \sin(5x)}{5} - \frac{1}{5} \cos(5x) = \frac{5x \sin(5x) + \cos(5x)}{25} + C$$

b) $\int_0^\pi x \sin x dx = x \cos(x) - \int \cos(x) dx = x \cos(x) - \sin(x)$

$$[\pi \cos(\pi) - \sin(\pi)] - [0 \cos(0) - \sin(0)] = -\pi \text{ u.a.}$$

12/04/20

S T Q Q S S D

⑥ Calcule o integral com TFC:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx = \left[\sin(x) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \sin\left(\frac{\pi}{2}\right) - \sin\left(-\frac{\pi}{2}\right) = 1 - (-1) = \boxed{2 \text{ u.a.}}$$